

A Genetic Algorithm-Based Optimization Approach for Disassembly Tool Selections toward Electric Vehicle Battery Disassembly

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Abstract—The electric vehicle battery of new energy vehicles has ushered in the first batch of decommissioning. In order to reduce its impact on the environment, the recovery and disassembly of the electric vehicle battery has received extensive attention. In view of the problems that the disassembly of new electric vehicle batteries still adopts manual operation and the robot cannot complete the disassembly task independently, and they need to select appropriate tools during the disassembly process, a genetic algorithm-based evaluation method for the selection of tools in the recycling and disassembly of electric vehicle batteries is proposed. The optimization model of disassembly tools is established, and then the proposed model is further studied and prospected.

Keywords—EV battery, disassembly tool, optimization approach, genetic algorithm

I. INTRODUCTION

Due to the continuous deterioration of climate and environment, as well as the requirements of relevant climate policies, emission reduction, green and environmental protection, the number of electric vehicles in various countries is increasing. According to the survey, by the end of 2021, the number of electric vehicles in China has reached 6.4 million. Then, the increase in the number of electric vehicles also means the increase of electric vehicle batteries, and also means the increase in the number of scrapped batteries. The first batch of electric vehicle batteries have been retired. According to the survey, the number of retired electric vehicle batteries in China in 2021 has reached 252000 tons. Waste electric vehicle batteries contain cobalt, lithium, nickel, manganese and other metal materials, which have high recycling value. At the same time, there are also harmful substances in the batteries that will pollute the environment. Therefore, it is a great challenge for the recycling of waste batteries [1-2].

The first step in the recovery process of electric vehicle battery is to discharge the battery. After discharging, the battery is disassembled to the battery core, and then a series of operations such as crushing, screening and separation, fine grinding are carried out to obtain high-value metal materials.

Therefore, the disassembly of waste batteries is an extremely critical step in battery recycling. However, at present, the field of electric vehicle battery disassembly in our country is still in the development stage, and it mainly relies on manual disassembly, which is inefficient. In addition, there are many dangerous operations in the disassembly process. At this time, robots are required to complete the disassembly, but it is not flexible to rely solely on robots for disassembly. Therefore, at present, most of the research at home and abroad rely on the cooperative operation of human and robot to complete the disassembly process. In the process of human collaborative disassembly, scholars at home and abroad have proposed many methods for disassembly sequence planning. Disassembly sequence refers to the optimal disassembly sequence for the electric vehicle battery disassembly process under certain constraints. Generally, the disassembly mode is determined first, then the disassembly information model is proposed, and then the objective function is given. Generally, the disassembly time, equipment cost and labor cost are taken as evaluation indicators, and finally the appropriate algorithm is selected to solve the problem, and the optimal disassembly sequence is obtained. For example, Yu et al. [3] proposed a task planning method for disassembly of automobile electric vehicle battery with destructive rules. By establishing a disassembly sequence model, reasonable disassembly planning can be carried out for parts with different connection modes. Zhou et al. [4] reviewed the existing main disassembly sequence planning methods, and also looked forward to the future research on disassembly sequence planning. Mitrouchev et al. [5] proposed a method of selective disassembly sequence based on virtual reality technology by analyzing the damage degree of workers in the process of work. In order to solve the unpredictable factors in the disassembly process, Tian et al. [6] adopted the hybrid algorithm to solve the proposed model. Xu et al. [7] used the discrete bee algorithm to solve the disassembly sequence planning, and used the mdba Pareto method to solve the optimal solution. However, few people have conducted cognitive evaluation research on the electric vehicle battery pack human collaborative disassembly tool. At present, there are many kinds of electric vehicle battery packs in the market,

and each battery manufacturer does not have a unified packaging standard. As a result, when facing the specific disassembly process, the disassembly tools involved in the human-machine cooperative operation are often selected and used by the operators. This greatly reduces the standardization and consistency of the disassembly process. In view of the change of disassembly tools caused by the change of disassembly projects, it is necessary to establish a unified and standardized cognitive evaluation system to establish a foundation for the upgrading of man-machine collaborative operation to production line automation.

This paper intends to propose an evaluation framework for tool selection in the process of electric vehicle battery disassembly to solve the difficulties of consistency and standardization of disassembly tools.

II. RESEARCH STATUS OF HUMAN-MACHINE COOPERATION

At present, the research on disassembling electric vehicle battery at home and abroad focuses on human-machine cooperation. Human-machine cooperation is to combine the advantages of human and robot to achieve an efficient work process. Man-machine cooperative disassembly refers to the disassembly of electric vehicle battery into individual parts in the way of joint mistake between human and robot under certain constraints. Lu et al. [8] studied the automatic disassembly platform of lithium-ion batteries, and used sensing and machine learning methods to protect the harm of lithium-ion batteries to human body. Christoph et al. [9] showed that it is almost impossible to disassemble the electric vehicle battery with a high degree of automation, and the robot assisted method must be adopted. At the same time, the safety of human body when people and robots work at the same time should be considered. Kai et al. [10] showed that artificial intelligence has limitations and complexity at present, and human robot cooperation is considered as the most effective method to solve uncertain problems. Kathrin et al. [11-12] proposed a battery disassembly workstation, and in the process of disassembly, robots assist humans, such as robots help humans remove screws and bolts, and take the battery of Audi Q5 as an example to design tools for different disassembly steps. Kay et al. [13] proved that the best way to disassemble the lithium-ion battery is human robot cooperation. The robot can cut the battery pack, and the workers classify and remove the disassembled parts on the battery pack. Smith et al. [14] planned a new disassembly sequence, considering various uncertain factors in the disassembly process to determine whether the parts are disassembled manually or by robots, so as to achieve the optimal disassembly sequence. Chen et al. [15] proposed a hybrid workstation, which can assist manual personnel to complete tasks such as loosening during disassembly, and control the safe distance between the robot and people to ensure personal safety. Most of the current research focuses on the specific implementation of human-machine collaborative operation and the design of different disassembly tools. Few people have established rigorous and reasonable cognitive evaluation methods and complete systems for the selection of tools in the process of human-machine collaborative disassembly. In the process of disassembly, man-machine faces the decision of tool selection. Establishing a cognitive evaluation system for tool selection in advance is

conducive to improving the automation level of the production line, greatly improving the disassembly efficiency and reducing the labor intensity of man-machine cooperative operation.

III. DISASSEMBLY PROCESS AND TOOL SELECTION

With the continuous increase of the total installed capacity of electric vehicle batteries, the first peak period of electric vehicle battery retirement has been reached. Because different automobile electric vehicle battery manufacturers have their own unique packaging structures from the perspective of their own intellectual property protection, they extract commonalities from different electric vehicle battery packaging structures to standardize the disassembly process. For the mainstream electric vehicle battery in the market, a unified disassembly process is summarized, as shown in Figure 1.

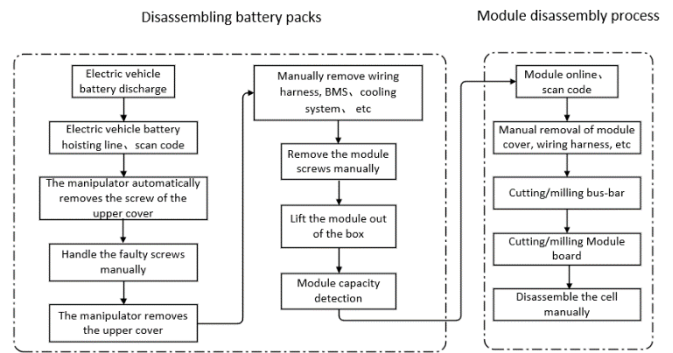


Fig. 1. Disassembly flow chart of electric vehicle battery.

As can be seen from Fig. 1, the tools involved in each process are different, and the first element of the selection of evaluation tools must be the connection mode. Fig. 2 shows the disassembly tools commonly used in the disassembly process of electric vehicle batteries.

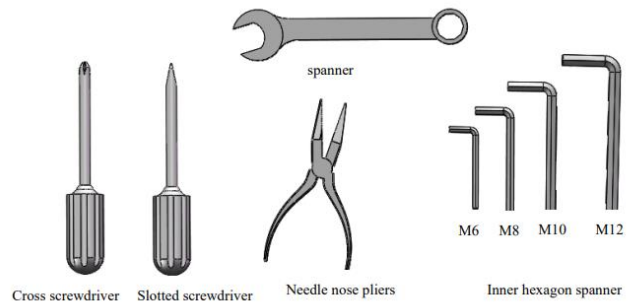


Fig. 2. Common disassembly tools.

Table 1 summarizes the specific connection methods, disassembly methods and selected tools involved in a certain type of electric vehicle battery. It can be summarized in three ways:

TABLE I. INFORMATION ABOUT THE ELECTRIC VEHICLE BATTERY REMOVAL TOOL OF A CERTAIN MODEL

Connection mode	Disassembly method	Disassembly position	Select tools
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Threaded connection	Automatic/non destructive	Battery upper screw module	pack cover and screw	UR Robot
	Manual/non destructive			Electric screwdriver, wrench
	Semi automatic/destructive	Faulty screw		Manipulator, Electric screwdriver, wrench
Glue connection	Manual/non destructive	Harness,BMS, cooling system, etc		Hand
	Semi automatic/non-destructive	Battery module and battery lower cover		Degumming agent
	Automatic/destructive	Module bar	bus	Milling robot
welding	Automatic/destructive	Module side plate		Milling robot

1) *Threaded connection*: Screw connection is the most common connection form of electric vehicle battery, mainly focusing on the connection between the upper end cover of the battery pack and the lower end cover of the battery pack; BMS, high-pressure box, cooling system, module and lower end cover. The removal of threaded connection is automatically disassembled by ur robot. Although the threaded connection is a standard connection, there are also screw connections of different series sizes that need to switch different types of screwdrivers. Therefore, according to the evaluation of the switch selection of the screwdriver during the disassembly process.

2) *Glue connection*: Glue connection is also a more common connection form in electric vehicle batteries, which is concentrated between the battery cells packaged into modules, modules and battery pack end caps. Most of the current treatment methods for glue connection are manual glue removal. However, due to the narrow position space of the parts connected with glue, the tools are difficult to reach, and the use of glue removal agent will lead to safety hazards, so the man-machine cooperation method is adopted. However, in the process of disassembly, there is a decision-making problem of how to select tools and models according to the location of the adhesive connecting parts of the battery pack. Therefore, it is necessary to select and evaluate the degumming tools according to the location and size of the adhesive connecting parts.

3) *Welding*: Welding is mainly concentrated in the fixed position of the bus bar inside the module. Due to the irreversible destructive disassembly, most of the current use of cutting technology and milling to disassemble the busbar. The bus bar is faced with local high temperature in the milling process, so the milling equipment is faced with the problem of adjusting and evaluating the rotation speed according to the real-time monitoring data of the temperature sensor.

In addition to establishing a cognitive evaluation system for tool selection according to different connection modes, it is also necessary to know that the whole disassembly production line is a whole design, so it is necessary to establish a cognitive evaluation system for overall human-machine collaborative

operation to improve the automatic production level and disassembly efficiency of the production line and standardize the tool and equipment management system.

IV. CONSTRUCTION OF DISASSEMBLY TOOL INFORMATION MODEL BASED ON GENETIC ALGORITHM

Human machine cooperation can realize the diversity of disassembly capabilities, and disassembly personnel and robots are complementary in the disassembly process. Because there are many uncertain factors in the disassembly process, for example, the structure of the electric vehicle battery is not certain, non-standard, unknown uncertain factors such as the type, quantity, size, and spatial constraint relationship of components, etc., and complete automation cannot be realized, so manual labor needs to be added. In the process of disassembly, the robot disassembles according to the fixed route according to the pre-set procedure. The disassembly accuracy is high, and the fatigue degree during manual operation can be solved. However, the ability of the robot to solve unexpected problems is limited, so it is necessary to add operators in the disassembly process. In the disassembly process, the selection of tools is the primary consideration. The disassembly tools have an important impact on reducing the disassembly time of electric vehicle batteries, improving the disassembly efficiency and reducing the fatigue of operators. According to the accessibility of the target disassembled parts, try to select efficient disassembly tools. For example, for the disassembly of a specific bolt, according to the swing angle of the bolt, since the operation space of the wrench is limited, the appropriate wrench shall be selected. The rotating wrench shall be selected as much as possible to improve the disassembly efficiency. For the disassembly of welded fixed parts, it is necessary to adopt appropriate methods to destructively disassemble them. To sum up, it is of great significance to select appropriate disassembly tools for improving disassembly efficiency. This paper proposes to establish a disassembly tool model based on genetic algorithm and conduct a cognitive evaluation system for disassembly tool selection.

The principle of genetic algorithm is survival of the fittest, which is a probability search algorithm based on the natural phenomena of living things in the natural environment. First, starting from the random mother chromosome, according to the selected fitness function, screening is carried out through replication, crossover and variation. The individuals with high fitness are left to form a new population. The new group has the information of the previous generation, but it is better than the previous generation. After screening again and again, until the appropriate group is selected.

Before cross mutation of each chromosome, it is first necessary to calculate its fitness value and determine whether it will be inherited to the next generation according to compare with the fitness value. The greater the fitness value, the greater the probability that an individual will be passed on to the next generation. The selection operation is to pass on the individuals with high current fitness to the next generation. The fitness value of the individual was then recalculated and then compared with the average fitness value of the population. Individuals with higher adaptability than the average adaptability correspond to lower cross probability to protect

individuals from entering the next generation; Individuals with lower than average fitness correspond to higher mutation probability, thus eliminating individuals:

$$P_c = \begin{cases} \frac{k_1(f_{\max}-f)}{f_{\max}-f_{\text{avg}}}, & f \geq f_{\text{avg}} \\ k_2, & f < f_{\text{avg}} \end{cases} \quad (1)$$

$$P_m = \begin{cases} \frac{k_3(f_{\max}-f')}{f_{\max}-f_{\text{avg}}}, & f' \geq f_{\text{avg}} \\ k_4, & f' < f_{\text{avg}} \end{cases} \quad (2)$$

Where, $f_{\max}$ refers to the maximum fitness value in the population; f_{avg} refers to the average fitness value of the population; f refers to the larger fitness value of the two to be crossed; f' refers to the fitness value of the individual to be mutated, k_1 、 k_2 、 k_3 and k_4 is a constant.

According to the basic principle of genetic algorithm, the flow chart of the selection of disassembly tools for electric vehicle batteries as shown in Fig. 3 is established to select appropriate disassembly tools in different disassembly processes.

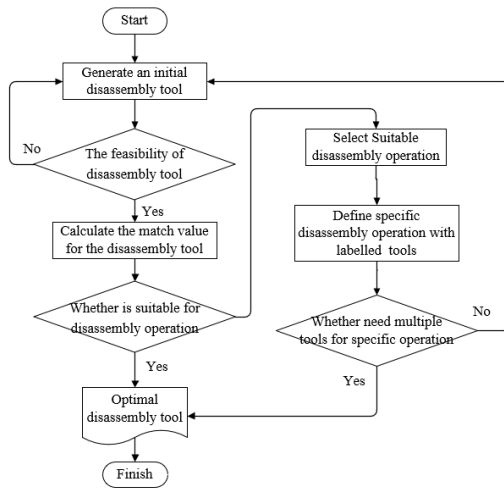


Fig. 3. Flow chart of genetic algorithm.

First, number the tools that may be used, as shown in Table 2. Then evaluate the feasibility of the disassembly tool. If it is not feasible, regenerate the disassembly tool. If it is feasible, calculate the matching value of the disassembly tool, and judge whether it conforms to the disassembly operation according to the size of the matching value, so as to form the optimal disassembly tool. If it does not conform to the disassembly operation, select the appropriate disassembly operation, define the specific disassembly operation with a label, and then judge whether the specific operation requires multiple tools. If not, regenerate the original disassembly tool. If yes, generate the best disassembly tool.

TABLE II. DISASSEMBLY TOOL NUMBER INFORMATION

Disassembly tools	Number
Electric screwdriver	R_1
Wrench	R_2

Hand	R_3
UR robot	R_4
Degumming agent	R_5
Milling robot	R_6
Needle nose pliers	R_7
Shovel	R_8

During the whole battery pack disassembly process, the electric screwdriver arranges the disassembly sequence according to the battery pack structure. While disassembling various parts, different tools need to be selected. Some parts can be disassembled faster by manual, while some complex parts can be disassembled faster by robot assistance, or for parts that are difficult to disassemble, destructive disassembly is dangerous, so robot must be used for disassembly, as shown in Table 3, Create the necessary tools according to the disassembly procedure.

TABLE III. DISASSEMBLY PROCEDURES AND NECESSARY TOOLS

Disassembly step No	Disassembly steps	Necessary tools
1	Remove the upper cover screws	R_4 、 R_1
2	Check and remove faulty screws	R_1 、 R_2
3	Remove the upper end cap	R_3
4	Remove harness/BMS/cooling water pipe, etc	R_7 、 R_3
5	Module debonding	R_5
6	Remove the lower cover of the battery pack	R_3
7	Remove the module upper cover screws	R_4 、 R_1
8	Remove the module upper cover screws	R_7 、 R_3
9	Remove the module upper cover screws	R_6
10	Remove the module side plate	R_6
11	Remove the module side plate	R_8 、 R_3

In this paper, disassembly time, disassembly cost and disassembly difficulty are taken as evaluation indexes for selection of man-machine collaborative disassembly tools to select appropriate disassembly tools.

(1) Disassembly time

Reducing disassembly time is a problem that must be solved in engineering. Disassembly time is directly related to the overall disassembly efficiency. According to different disassembly steps, compare the operation time of manual and robot, and the disassembly time of man-machine cooperation, so as to select the minimum disassembly time.

(2) Disassembly difficulty

In the face of different disassembly components, the difficulty of manual disassembly is different from that of robot disassembly, and some components must be disassembled by robot. Therefore, the difficulty of disassembly is selected as the evaluation index of tool selection in the disassembly process.

(3) Dismantling cost

For the company, it is necessary to reduce the cost of disassembly. The reduction of disassembly cost can bring more benefits to the enterprise. The labor cost and robot cost are all problems to be solved. The number of disassembly tools and the type of disassembly tools to choose are major problems.

V. SUMMARY AND PROSPECT

Faced with the characteristics of different types and structures of retired electric vehicle batteries, combined with the shortcomings of manual disassembly and automatic disassembly, the man-machine collaborative disassembly improves the overall disassembly efficiency. Considering the characteristics of the disassembly ability of human and robot, there are many uncertain factors in the disassembly process, a common method and steps for the disassembly of electric vehicle batteries are summarized, and appropriate tools are selected for different disassembly parts. A cognitive evaluation system of tool selection based on genetic algorithm is proposed. Genetic algorithm is a random global search optimization method. For the disassembly of electric vehicles, people often use it to study the disassembly sequence planning, but there is no research on the selection of tools in the disassembly process. Selecting appropriate disassembly tools is also an important research topic. In the face of different disassembly processes and disassembly under uncertain conditions, how to select appropriate disassembly tools to reduce disassembly time, reduce disassembly cost, reduce manual fatigue and reduce disassembly difficulty is an important issue. In this paper, genetic algorithm is used to evaluate disassembly tools, and different evaluation systems can be used to optimize them in the future. In a word, the research of disassembly tool selection is a complicated and necessary process.

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